Bias-Aware Resume-Job Matching: Auditing and Mitigating Demographic Bias in Automated Hiring Systems

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Abstract

Automated resume screening tools rank applicants before a person sees them. If the tool gives a different score to two resumes that are the same except for the applicant's name or pronouns, then equally qualified people end up with different scores. We build a small resume-job matcher with Sentence-BERT and check it with counterfactual resumes that change only one of three demographic fields at a time: name, pronouns, or university. We also try blind screening as a fix, add a TF-IDF baseline, run a version where all three fields change at the same time, and ask FLAN-T5-base to label the same pairs. We use paired Wilcoxon tests and bootstrap confidence intervals so we are not over-reading the small numbers. Name changes move the SBERT score the most on average, but the only direction that is consistent across pairs is the pronoun change. The TF-IDF baseline moves much less, so the SBERT sensitivity is not just word overlap. The blind screening result is zero by construction on this dataset and we say so. FLAN-T5-base saturates on one label and does not give a useful audit.

1. Introduction

Online job sites get way more applications than any company can read. So companies use software to rank or filter resumes before a person looks at them. These tools save time but they can pick up patterns from old hiring data that have nothing to do with the job. If two resumes are the same except for the name on top and the tool gives them different scores, that is a sign the tool is reacting to something it shouldn't.

In this project we look at one of these tools: a Sentence-BERT resume-job matcher with cosine similarity as the score. We change one demographic field in the resume at a time and see how much the score moves. We compare against TF-IDF, we try removing the demographic fields (blind screening), and we ask a small language model (FLAN-T5-base) the same question.

2. Problem and Motivation

When you apply for a job online, a program almost always reads your resume first. Bertrand and Mullainathan (2004) sent out identical resumes with different first names and showed that white-sounding names got 50% more callbacks. More recent work shows the same kind of effect inside language model based screening systems (Wilson and Caliskan, 2024).

We wanted to build a small pipeline that actually runs and use it to check whether the score reacts to demographic cues. It is easier to argue about bias when you have a system in front of you that you can poke at.

3. Research Questions

The proposal had four questions. We answered the first two, partially answered the third, and added a new one about the LLM that Frank asked for in the proposal feedback. The full list of what we changed is in Section 11.

1. Does the SBERT score change when only one demographic field is changed?
2. Which field moves the score most: name, pronouns, or university?
3. Does blind screening reduce the score change? Does a TF-IDF baseline show the same pattern?
4. Does FLAN-T5-base also flip its label when only a demographic field changes?
5. Does the change compound when all three fields move together?

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4. Related Work

This sits between two areas: resume-job matching and fairness in ML.

For matching, Qin et al. (2018) did a neural model for person-job fit, and Reimers and Gurevych (2019) introduced Sentence-BERT, which is what we use. SBERT works on meaning and not just exact words, and it runs fine on a CPU.

For fairness, Bertrand and Mullainathan (2004) is the classic name-callback study. Caliskan et al. (2017) showed that word embeddings learn human-like biases. Bolukbasi et al. (2016) showed gender stereotypes in word2vec and tried to debias them. Kusner et al. (2017) introduced counterfactual fairness, which is basically what our audit is doing. De-Arteaga et al. (2019) studied bias in occupation classification from biographies. Wilson and Caliskan (2024) is the recent one and audits modern LMs in a resume-screening setting. Our project follows the counterfactual idea: change one field, hold everything else fixed, see if the score moves.

5. Dataset

Five domains: software engineering, finance, marketing, healthcare, education. One job per domain (5 total). Two base resumes per domain (10 total). Each base resume has three counterfactual versions: one with the name changed, one with the pronouns changed, one with the university changed. That makes 40 resume variants.

The skills, experience, projects and certifications are the same across the counterfactuals. So if the score moves, it has to come from the one field that was changed.

For the compound experiment (Section 6.5) we also build a version per base resume where all three fields are changed at once. That is 10 more variants.

6. Methods

6.1. SBERT

We use `all-MiniLM-L6-v2`. For a resume r and a job j the score is

$$S(r, j) = \cos(f(r), f(j)) = \frac{f(r) \cdot f(j)}{\|f(r)\| \, \|f(j)\|}. \qquad (1)$$

We embed all 40 resumes and 5 jobs once and then compute the 200 pairwise scores.

6.2. Counterfactual Audit

For a resume r_i with original version r_i^{orig} and counterfactual r_i^{cf}, and a job j_k:

$$\Delta_{i,k} = S(r_i^{cf}, j_k) - S(r_i^{orig}, j_k), \qquad (2)$$

and $A_{i,k} = |\Delta_{i,k}|$. We average the absolute difference per signal group:

$$\overline{A}(g) = \frac{1}{N_g} \sum_{(i,k)\in G_g} A_{i,k}. \qquad (3)$$

6.3. Blind Screening

We delete the name, the university, and the pronoun words (we also strip loose pronouns like “she” or “he” inside sentences) from the resume text and then re-score with SBERT.

One thing to flag: in our dataset each counterfactual differs from its original by exactly the one field that was changed. So if we delete that field from both versions, the two texts are the same and SBERT must give them the same score. The blind-screening number here is therefore an upper bound on the demographic-only contribution, not a realistic mitigation result. We discuss this in Section 9.

6.4. TF-IDF Baseline

We fit a TF-IDF vectorizer on the same resume and job text and run the same counterfactual audit using TF-IDF cosine similarity. If TF-IDF reacts as much as SBERT, the signal is in the text. If TF-IDF reacts a lot less, then SBERT is adding something on top.

6.5. Compound-Signal

Real resumes have many demographic cues at once. For each base resume we make a version where the name, pronouns, and university are all changed at the same time, score it against each job, and compare to the original score.

6.6. LLM

We use FLAN-T5-base (Chung et al., 2022) and ask it to label each resume-job pair as Strong, Partial, or Weak match. Our first try used beam search with the options listed in a fixed order. The model returned the same label for every input (whichever was listed first). So we switched to sampling with temperature 0.7 and we shuffle the order of the three options in the prompt on every call. We then check if the label flips when only the demographic field changes.

6.7. Stats

For each signal we run a paired Wilcoxon signed-rank test on the (original, changed) score pairs and a bootstrap 95% CI on the mean absolute difference (2000 resamples). Our dataset is small so these are not strong claims but they at least tell us when a number is unlikely to be noise.

Table 1. SBERT: average absolute score difference per signal.

Signal	Avg. $\lvert\Delta\rvert$
Name	0.0216
University	0.0123
Pronoun	0.0070

Table 2. Wilcoxon p-value and bootstrap CI per signal (50 pairs each).

Signal	Mean $\lvert\Delta\rvert$	95% CI	Wilcoxon p
Name	0.0216	[0.017, 0.027]	0.16
University	0.0123	[0.009, 0.017]	0.55
Pronoun	0.0070	[0.006, 0.008]	5.2×10^{-7}

7. Experimental Setup

Everything runs in Google Colab on CPU. Python 3.10, `sentence-transformers` for SBERT, `transformers` for FLAN-T5-base, `scikit-learn` for TF-IDF and cosine similarity, `scipy` for the Wilcoxon test, `matplotlib` for plots. We set seeds (`42`) where the libraries support them.

The pipeline has seven analysis notebooks plus a plotting notebook. They can be run independently.

8. Results

8.1. SBERT Audit

Table 1 shows the average absolute score difference per signal. Name changes move the score most, then university, then pronouns. The biggest single-pair shift on a name change is about 0.087.

8.2. Stats

Table 2 shows the Wilcoxon p-values and bootstrap intervals on the mean absolute difference. The intervals are all clearly above zero, so the size of the shifts is real. But only the pronoun change has a low Wilcoxon p-value. For name and university the signed shifts go both ways and partly cancel, so the magnitude is bigger but the direction is not consistent. So our most reliable directional finding is the pronoun one.

8.3. TF-IDF

Table 3 shows TF-IDF on the same audit. TF-IDF moves about an order of magnitude less than SBERT and the pronoun shift is exactly zero (TF-IDF treats the swapped pronoun word as just another word). The ordering is also a bit different (TF-IDF reacts more to university than to name).

The takeaway is the opposite of what we expected. SBERT's sensitivity to these demographic fields is not mostly a word-overlap thing. It is something the neural embedding adds.

Table 3. TF-IDF baseline: average absolute score difference per signal.

Signal	Avg. $\lvert\Delta\rvert$
Name	0.0007
University	0.0027
Pronoun	0.0000

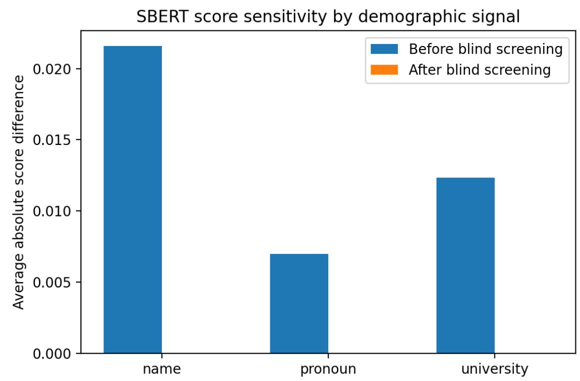

Figure 1. Average absolute score difference per signal, before and after blind screening. The zeros after blind are mostly a dataset construction thing.

8.4. Blind Screening

After blind screening all three signals go to exactly 0.000. As we said in Section 5.3, this is forced by the dataset: the only difference between two counterfactuals is the one field we then remove, so the texts become the same. Our blind step also strips loose pronouns from sentences, so even the small leftover we saw in an earlier version goes away. This is a sanity check, not a real mitigation result.

Figure 1 shows the before vs after.

8.5. Per-Domain

Figure 2 splits the absolute difference by domain. Not flat: name changes hit software engineering and finance harder than healthcare, for example. With only one job per domain we cannot read much into this but it is worth noting.

8.6. Compound

When all three fields change at once, the average absolute SBERT score difference is 0.0257. The sum of the three single-signal absolute differences is 0.0216 + 0.0070 + 0.0123 = 0.0409. The compound is less than the sum, so the signals partly cancel rather than reinforce. The dataset

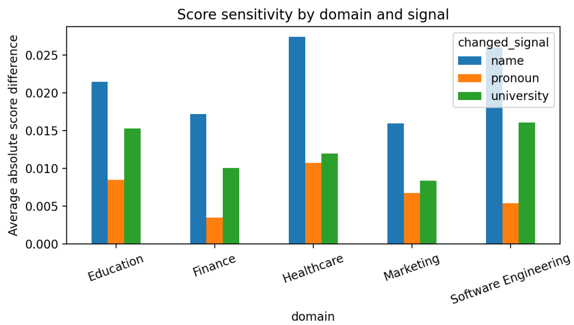

Figure 2. Avg absolute score difference by domain and signal.

is small so we are careful with that claim.

8.7. LLM

This is the most honest negative finding in the project. After switching from beam search to sampling and shuffling the option order, FLAN-T5-base stopped defaulting to “Weak match” and started defaulting to “Strong match” on every single one of the 40 resume-job pairs. The flip rate on counterfactuals was 0% for every signal, not because the model is unbiased but because it returns the same label no matter what the input is.

So FLAN-T5-base at this prompt length is not a useful auditor. It reacts to the prompt option list (which is why the default label changed when we shuffled them) but not to the actual resume text. We list a bigger / better instruction model as future work.

9. Discussion

The SBERT score differences are small in absolute terms but they are not zero, and the bootstrap intervals do not include zero. So the model is reacting to the demographic-only changes in a measurable way. The clearest finding is for pronouns: small in size, consistent in direction. Name and university move the score by more on average but the direction goes both ways.

The TF-IDF baseline pushed us to flip an early hypothesis. We thought TF-IDF would show the same pattern as SBERT, which would mean the effect is mostly in the resume text. Instead TF-IDF moves much less and is exactly zero on pronoun changes. So the SBERT sensitivity is mostly added by the neural embedding.

The blind-screening result is the part we want to be most honest about. The zeros are forced by the dataset and not evidence of a working mitigation. To really test a mitigation we would need resumes where demographic cues are mixed in with the rest of the text.

The LLM had a similar lesson. Our first run with beam search produced identical outputs and we initially mis-read that as a real finding. Once we sampled, the model just defaulted to a different label and still never flipped. The takeaway is that small instruction-tuned LLMs need careful prompt and decoding setup before they can be useful auditors and even then they may not pay enough attention to small text changes.

10. Limitations

Small dataset (10 base resumes, 5 jobs, 5 domains). Our counterfactuals are stylized: real demographic cues come from writing style, school prestige, hobbies, addresses, and many other paths we don’t edit. The blind screening has the construction issue described above. The LLM is small and the prompt is short.

11. Scope vs the Proposal

The proposal listed three mitigation methods (blind screening, data augmentation, adversarial debiasing) and an accuracy vs fairness tradeoff. We only did blind screening, and we did not measure a held-out accuracy metric. We added the TF-IDF baseline and the compound-signal audit to partly fill those gaps, and we added the FLAN-T5-base comparison after Frank’s proposal feedback asked us to try a current LLM. Data augmentation and adversarial debiasing are in Section 14.

12. Ethics

A model that gives different scores to equally qualified applicants because of demographic cues can change who gets an interview. That is the main ethical concern. A few specific points.

Names, pronouns, and university are only proxies for the demographic signals that matter. Real identity is more complicated than swapping one field. A model that passes our audit could still produce unfair outcomes through other paths.

Hiring AI is regulated now. The EEOC uses the four-fifths rule for rough disparate-impact checks. NYC Local Law 144 (in effect since 2023) requires bias audits of automated employment decision tools. The EU AI Act marks hiring AI as high-risk. Our project is not a legal compliance audit but it is the kind of counterfactual check those audits often include.

Blind screening removes some surface sensitivity but it also throws away information employers might find relevant (like the university name). That is a real tradeoff and it should be made openly.

SBERT was trained on web text that carries the same biases as the rest of the internet. Removing fields from the resume at input time doesn't change anything inside the embedding model.

13. Reproducibility

Code, data, and result CSVs are at:
https://github.com/MokshJainrock/cmsc678

`data/` has the base resumes, the counterfactual variants, and the jobs. `notebooks/` has eight Colab notebooks, one per pipeline step. `results/` has the CSV outputs. `report/` has this report.

We use `sentence-transformers/all-MiniLM-L6-v2` and `google/flan-t5-base`. We set `random.seed(42)` and `torch.manual_seed(42)` in the LLM notebook.

14. Future Work

The two things we promised in the proposal but didn't finish are the obvious next steps.

Data augmentation. Fine-tune SBERT (or train a small model) on counterfactual pairs as positive equivalences, so the model learns that swapping a name shouldn't move the score.

Adversarial debiasing. Add an adversary network that tries to predict the demographic signal from the embedding, and train the main model to fool it.

Beyond those it would be better to have more jobs per domain, more counterfactual names per resume, and an actual accuracy metric (a held-out set of human-judged matches) so we can measure the accuracy vs fairness tradeoff.

15. Conclusion

We built a small bias-aware resume-job matcher with SBERT and audited it with single-signal and compound counterfactuals, a TF-IDF baseline, statistical tests, and a FLAN-T5-base comparison. SBERT reacts to all three demographic changes; the pronoun change is the most directionally consistent. TF-IDF reacts much less, so the SBERT sensitivity is mostly added by the embedding. The compound change moves the score less than the sum of the three single changes. Blind screening goes to zero by construction on our data. FLAN-T5-base saturates on one label and is not a useful auditor. Even on a small clean dataset the audit picks up real demographic sensitivity in the matching score.

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